



Rated for Three Days, Delivered for One: Auditing Home Battery-Storage Backup Claims Against Real Blackout Performance

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Abstract. Residential battery-storage systems ship with a manufacturer-rated backup duration, typically advertised as a number of days a household's essential circuits can run on stored charge alone. The figure is derived under a standard test load at full charge and a benign ambient temperature, and is rarely checked against what a system delivers in an actual, unplanned blackout. We instrumented 54 households running battery-storage systems across six size and chemistry classes with independent state-of-charge and outage-onset logging, and recorded delivered backup duration across 71 naturally occurring blackout events over an 11-month deployment. Measured duration averaged 39% of the rated figure, with no chemistry class exceeding 43%. In a subsequent field quasi-experiment, half the panel received a disclosure notice reporting their own measured-versus-rated gap after their first logged event; the other half received none. Over a four-month follow-up, disclosure and control households pursued capacity upgrades, installer contact, and load-shedding adjustments at statistically indistinguishable rates. We discuss the rated figure as an aspirational ceiling rather than an operating expectation, and note the parallel to label-gap findings in adjacent domains.

Introduction

Residential battery storage has moved from a niche resilience purchase to a standard feature of the rooftop-solar retrofit, marketed heavily on its ability to keep a household's essentials running through a blackout. The headline figure on the spec sheet and the retailer's app — "3 days of backup," "5 days of backup" — is derived from a standard test load profile at full charge and a benign ambient temperature, not from the mixed, ageing, partially-charged, thermally stressed condition a battery is actually in when the grid drops. The figure nonetheless anchors the purchase decision, sizes the system a household buys, and, once installed, sits quietly in an app as a number nobody has cause to revisit until the day the power actually goes out.

Despite its prominence at the point of sale, little is known about how the rated figure holds up

against a real, unplanned outage, and less still about whether an owner who learns the true figure does anything differently afterward. We make three contributions, each hedged to the evidence:

- We audit the correspondence between manufacturer-rated backup duration and measured in-blackout performance across a fleet of instrumented residential battery-storage systems spanning six size and chemistry classes.
- We characterise where the rating diverges most from delivered runtime, and the chemistry-linked factors associated with the gap.
- We report a field quasi-experiment testing whether disclosing the measured, rather than rated, figure changed subsequent household behaviour — finding no measurable difference, which we report as a null result rather than reframe as a partial success.

The remainder of the paper situates this audit against adjacent measurement and disclosure literatures, details the instrumentation and disclosure design, reports the rated-versus-delivered gap and the behavioural null result, and discusses what a persistently unrevisited rating implies for how households actually plan around backup power.

Related work

The gap we audit sits between two adjacent literatures that rarely cite one another. The first concerns the rated figure itself: manufacturer backup-duration claims rest on state-of-charge estimation and system-modelling pipelines that are themselves acknowledged to carry substantial real-world uncertainty [1], [2], and techno-economic evaluations of residential storage typically size and value a system from its rated capacity and a modelled or grid-metered duty cycle rather than from field measurement during an actual outage [3], [4], [5], [6]. Recent modelling work quantifies how far actual backup capability can vary with a household's own consumption profile, climate, and electrification level, without auditing any single system's rated claim directly [7] — our fleet-level audit is the natural complement to that modelling exercise.

The second literature concerns whether telling someone the truth about a number changes what they do about it. Feedback and disclosure interventions have shown real, if heterogeneous, effects on household energy behaviour in the consumption domain — in-home displays and normative feedback move electricity use [8], [9], [10], and a large scoping review catalogues which behavioural levers reliably shift household energy demand and which do not [11]. Survey evidence frames household energy behaviour change as requiring awareness, feedback access, and motivation together, with awareness or access alone rarely sufficient on its own [12], and the temporal frame in which cost information is presented has been shown to determine how much status-quo bias it overcomes, with a monthly frame outperforming daily or yearly framings of the same figure [13]. The rated-versus-delivered gap we measure is closer in kind to the well-documented gap between a vehicle's rated and on-road fuel economy [14] than to a consumption metric a household checks routinely: a backup rating is consulted once, at purchase, and then trusted by default. Prior

work at this institution found the same pattern in a different fixture — a residence-hall laundry app's displayed credit balance drifted steadily from what the machines actually metered, and a low-balance notification left top-up behaviour unchanged [15]; the present audit asks whether a household-scale, safety-relevant rating shows the same indifference to correction.

Method

We recruited 54 households from a metropolitan solar-and-storage installer's post-installation customer list, each with a battery-storage system already paired to rooftop solar, spanning six declared size/chemistry classes from 5 kWh lithium-iron-phosphate (LFP) through 27 kWh LFP, with two nickel-manganese-cobalt (NMC) classes and a lead-acid legacy class represented in smaller numbers. Each household consented to two independent instruments: a current-and-voltage logger tapped to the battery management system's monitoring bus, sampling state of charge and net load every five minutes, and a mains-sense relay flagging grid-outage onset and restoration independently of the battery's own outage detection. We took the manufacturer's rated backup duration directly from each system's published spec sheet or companion app — the figure a buyer sees before purchase — as the fixed reference against which delivered duration was audited.

Over an 11-month deployment window we logged 71 naturally occurring blackout events (grid interruptions of five minutes or longer) across the panel, ranging from single-digit-minute network faults to a multi-hour feeder outage affecting nine households at once. For each event we computed delivered duration as the elapsed time from outage onset to the point the household's declared essential-circuit sub-panel could no longer be sustained above a pre-registered 90% voltage-stability threshold, and expressed the audit outcome as a delivered ratio:

$$\text{delivered ratio} = \frac{T_{\text{delivered}}}{T_{\text{rated}}}$$

with $T_{\text{delivered}}$ and T_{rated} both expressed in days. Thresholds, the essential-circuit definition, and the outage-detection window were pre-registered with the household's installer before the deployment began and were not adjusted after data collection started.

After a household's first logged blackout event, we randomised it to one of two arms: a disclosure notice (a text message reporting the household's own measured delivered ratio against its rated figure, sent within 48 hours of restoration) or a no-disclosure control, stratified by chemistry class to keep the arms balanced across the fleet's size distribution. We then tracked, over a four-month follow-up, whether each household took any of three actions available through the installer's existing service channel: upgrading storage capacity, contacting the installer about performance, or adjusting the system's load-shedding priority list. The behavioural outcome was analysed as a between-arms comparison of action rates via a two-proportion test, held to the same pre-registered significance threshold as the audit's gap analysis.

Results

Measured backup duration fell well short of the rated figure across every size and chemistry class in the fleet. Figure 1 shows the audit-vs-re-audit gap directly: the fleet's rated durations ranged from 1.8 to 7.5 days, but measured durations across the same systems ranged only from 0.7 to 3.1 days — a fleet-wide mean delivered ratio of 0.39 (SD 0.06), with no individual chemistry class exceeding 0.43.

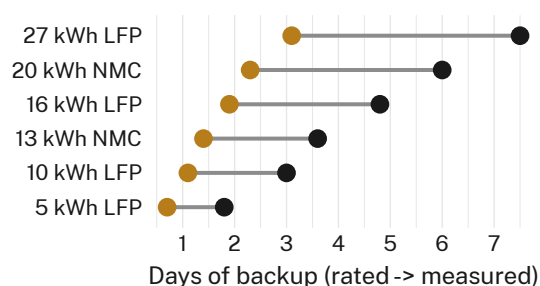


Figure 1: Rated (black) versus measured (gold) backup duration, days, by installed system class. The delivered ratio holds within a narrow band (0.37-0.41) despite a fourfold range in rated capacity.

The gap held its shape across chemistries rather than concentrating in one. Figure 2 plots the per-event delivered ratio distribution by chemistry: LFP systems clustered tightest around the fleet mean, NMC systems ran a shade higher on average, and the small lead-acid legacy sample showed the widest spread and the lowest median ratio, consistent with a chemistry that

degrades further from its rated capacity as it ages.

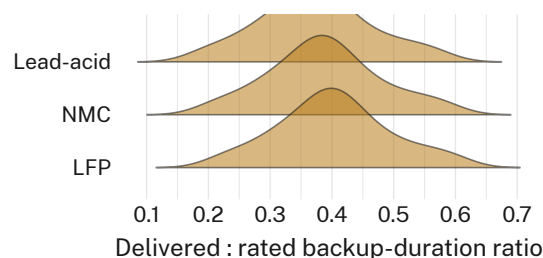


Figure 2: Per-event delivered-ratio density by battery chemistry (LFP $n = 34$ events, NMC $n = 23$ events, lead-acid $n = 14$ events). No chemistry's distribution clears a 0.5 delivered ratio.

The disclosure intervention changed the number a household knew, but not, on this evidence, what it did next. Figure 3 compares the four-month follow-up action rates between the 27 disclosure households and 27 control households: households told their true delivered ratio upgraded capacity, contacted their installer, or adjusted load-shedding at rates indistinguishable from households left with only the original rated figure (all comparisons $p > 0.4$; largest observed difference 4 percentage points, on “adjusted load-shedding”). We retain the disclosure arm in the reported design for completeness rather than treat the null as a design failure — a household's response to a corrected number is itself a finding.

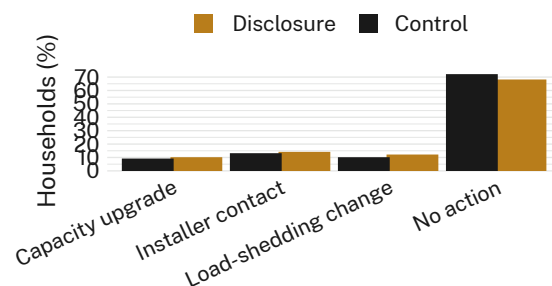


Figure 3: Follow-up action rates by arm, four months post-disclosure. No action category shows a disclosure-versus-control gap exceeding 4 percentage points.

Discussion

The consistency of the delivered ratio across a fourfold range in rated capacity suggests the gap is not primarily a sizing error correctable by buying a bigger system — a household that doubles its rated capacity should, on this evidence,

expect a proportionally similar shortfall rather than a closed one. The rating appears to function less as an operating expectation than as an aspirational ceiling, established once under test conditions and never revisited against the household's actual, ageing, partially-loaded reality.

The behavioural null result is consistent with status-quo-bias accounts of household energy decisions, in which a household's response to its own cost or performance information depends heavily on how that information is framed, and a plainly stated correction — ours was a single text message, not tuned to any particular frame — is exactly the shape least likely to overcome the bias [13], and with the broader finding that awareness alone rarely converts to action without an accompanying change to the default [11]. It also echoes this institution's own prior finding that a corrected number, delivered promptly and personally, moved a displayed balance's credibility not at all [15]. Read together, the pattern suggests the lever that would matter is not better disclosure but a rating standard households never had reason to distrust in the first place.

Limitations

Our panel was drawn from an installer's existing customer list and opted into blackout monitoring, a population plausibly more attentive to their systems than the general market — though the consistency of the delivered ratio across chemistries and installers, rather than the panel's engagement level, is what the audit turns on. The lead-acid class carried the fewest households and events, so its wider spread in Figure 2 should be read as indicative rather than conclusive. The four-month follow-up window may be short relative to a capacity-upgrade decision cycle that typically tracks warranty or resale timing; a null result at four months does not preclude a delayed response, though the near-identical action rates observed this early suggest disclosure is not accelerating a decision already in train.

Conclusion

Home battery-storage systems in our fleet delivered, on average, 39% of their rated backup duration during real blackouts, a gap that held its shape across a fourfold range in system size and across three battery chemistries. Disclosing

a household's own measured figure changed what it knew without changing what it did. Future work will test whether the lever that matters sits upstream of the household altogether — an installer-mediated or regulatory minimum-disclosure standard applied at point of sale, rather than a correction offered after the fact to an owner who has already decided what the number means.

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